Activity: Run Simple Linear Regression

Introduction

Simple linear regression is a statistical technique used to model the relationship between two variables. By assessing the direction and strength of this relationship, data professionals can identify patterns and transform data into valuable insights. This helps businesses make better predictions and data-driven decisions.

In this project, you are part of the analytics team at a fictional online electronics store that sells products such as smartphones, laptops, and electronic accessories. The company invests in different digital marketing channels, including Google Ads, YouTube advertising, and social media campaigns. Your task is to explore the relationship between Google Ads expenditure and sales revenue.

The dataset contains information about advertising expenditure across different digital marketing channels, campaign types, product categories, and the corresponding sales revenue. Using this information, you will investigate whether Google Ads expenditure is associated with changes in sales.

The findings will help the analytics team understand the relationship between advertising investment and sales revenue, supporting more informed decisions about future marketing activities.

In [ ]:
```
!pip install statsmodels
```

Step 1: Imports

In [14]:
```
# Import the relevant Python libraries and modules required for data analysis, visualization, and simple
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.formula.api import ols
import statsmodels.api as sm
```

The dataset provided is an Excel file named `Online_Electronics_Store_Marketing_Sales.xlsx`, which contains information about digital marketing expenditure across different advertising channels, along with the corresponding sales revenue. Assume that the numerical variables representing advertising expenditure and sales are expressed in thousands of Indian rupees (₹). The dataset includes information about Google Ads, YouTube advertising, social media spending, campaign types, and product categories.

Note: This is a fictional dataset created for educational purposes. The data is intended to help analyze the relationship between Google Ads expenditure and sales revenue using simple linear regression.

In [17]:
```
## Step 2: Load the dataset
data = pd.read_excel(r"C:\Users\LENOVO\Downloads\Online_Electronics_Store_Marketing_Sales.xlsx")
```

Step 2: Data exploration

In [19]:
```
# Display the first 10 rows of the data.
data.head(10)
```

Out[19]:

	YouTube_Ads_Spend	Google_Ads_Spend	Social_Media_Spend	Campaign_Type	Product_Category	Sales
0	47.25	38.34	13.43	Shopping	Smartphones	179.15
1	59.40	57.36	24.43	Search	Laptops	242.27
2	33.44	55.24	9.17	Performance Max	Accessories	205.24
3	3.83	59.49	3.09	Remarketing	Accessories	194.11
4	29.21	43.37	33.96	Search	Accessories	162.59
5	25.61	54.27	39.52	Search	Smartphones	229.71
6	17.56	105.13	36.86	Shopping	Smartphones	358.29
7	10.67	77.90	32.02	Performance Max	Smartphones	235.32
8	12.06	17.39	19.32	Remarketing	Accessories	122.94
9	23.83	75.44	58.91	Performance Max	Smartphones	283.13

In [24]:
```
# Display the shape of the dataset
data.shape
```

Out[24]:
```
(572, 6)
```

In [25]:
```
# Display the info of the dataset
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 6 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   YouTube_Ads_Spend   572 non-null    float64
 1   Google_Ads_Spend    571 non-null    float64
 2   Social_Media_Spend  572 non-null    float64
 3   Campaign_Type       571 non-null    str    
 4   Product_Category    572 non-null    str    
 5   Sales               571 non-null    float64
dtypes: float64(4), str(2)
memory usage: 26.9 KB
```

In [26]:
```
# Now describe the dataset
data.describe()
```

Out[26]:

	YouTube_Ads_Spend	Google_Ads_Spend	Social_Media_Spend	Sales
count	572.000000	571.000000	572.000000	571.000000
mean	31.887675	53.425236	33.343217	193.578266
std	22.177818	32.142630	22.146138	74.179549
min	2.000000	8.000000	3.000000	45.410000
25%	15.485000	29.630000	16.815000	140.820000
50%	27.120000	46.540000	28.490000	186.420000
75%	43.230000	71.290000	44.130000	235.520000
max	120.000000	180.000000	130.000000	517.250000

Dataset Overview

Size: 572 rows and 6 columns.

Features:

Numeric (float64): YouTube_Ads_Spend, Google_Ads_Spend, Social_Media_Spend, and Sales.

Categorical (str): Campaign_Type and Product_Category.

Data Check: Minor missing values exist in Google_Ads_Spend, Sales, and Campaign_Type (each has 571 out of 572 non-null entries).

Statistical Summary

Ad Spend:

Google Ads has the highest average investment (53.43), peaking at 180.00.

Social Media averages 33.34 (max 130.00) and YouTube Ads averages 31.89 (max 120.00).

Sales:

Average sales stand at 193.58, with half the records falling below 186.42 and a maximum peak of 517.25.

In [28]:
```
# Start with .isna() to get booleans indicating whether each value in the data is missing.
data.isna()
```

Out[28]:

	YouTube_Ads_Spend	Google_Ads_Spend	Social_Media_Spend	Campaign_Type	Product_Category	Sales
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
567	False	False	False	False	False	False
568	False	False	False	False	False	False
569	False	False	False	False	False	False
570	False	False	False	False	False	False
571	False	False	False	False	False	False

572 rows × 6 columns

In [32]:
```
# Use .any(axis=1) to get booleans indicating whether there are any missing values along the columns in

data.isna().any(axis=1)
```

Out[32]:
```
0      False
1      False
2      False
3      False
4      False
       ...
567    False
568    False
569    False
570    False
571    False
Length: 569, dtype: bool
```

How many rows containing missing values?

Based on the output(Length: 569, dtype: bool), there are 3 rows containing missing values out of the total 572 rows.

In [33]:
```
# Remove rows containing missing values
data = data.dropna(axis=0)
```

In [34]:
```
# Start with .isna() to get booleans indicating whether each value in the data is missing.
# Use .any(axis=1) to get booleans indicating whether there are any missing values along the columns in
# Use .sum() to get the number of rows that contain missing values

### YOUR CODE HERE ###
data.isna().any(axis=1).sum()
```

Out[34]:
```
np.int64(0)
```

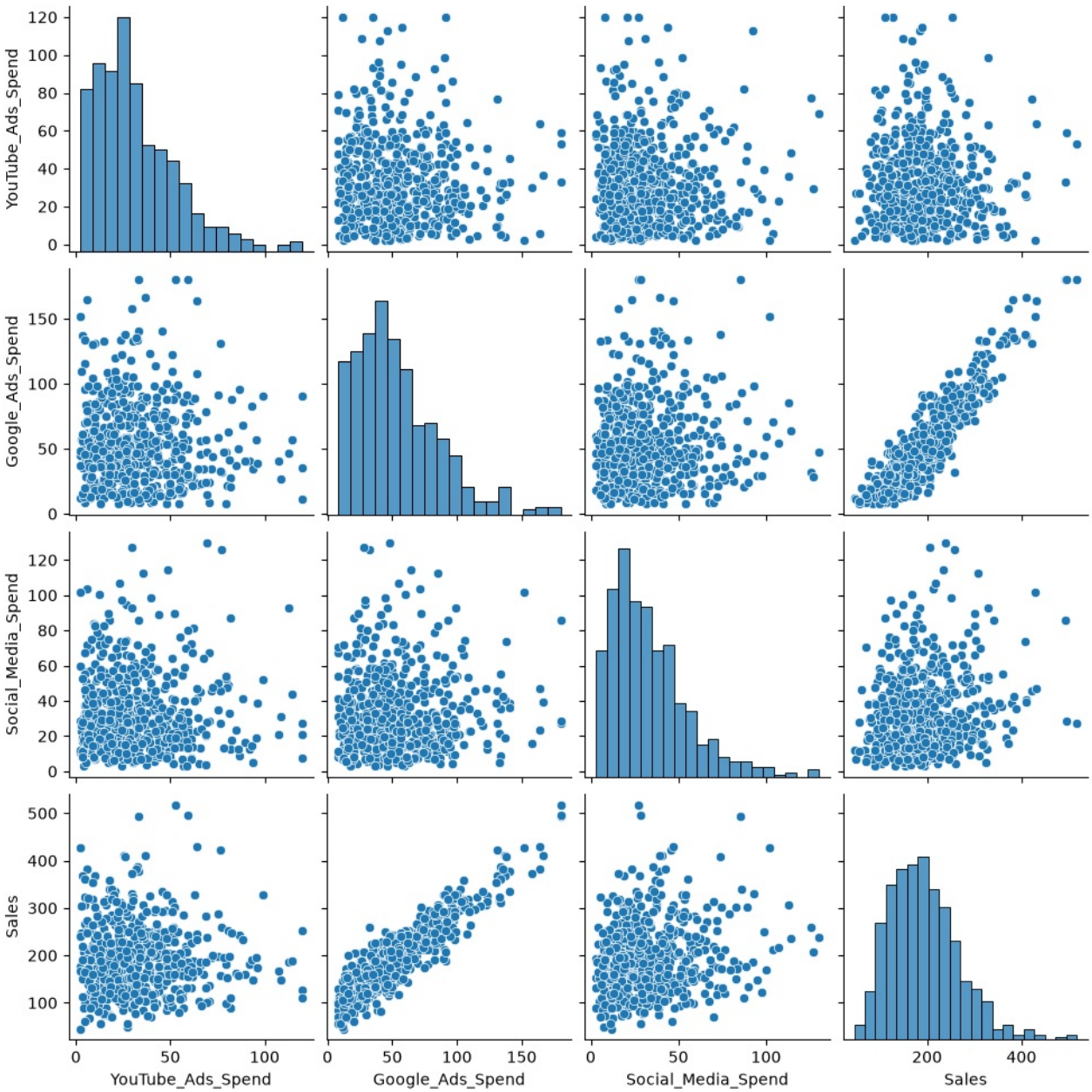
Step 3: Check Model Assumptions

The next step is to check the assumptions of linear regression. To explore the relationship between Google Ads expenditure and sales revenue, we will first visualize the pairwise relationships between the variables in the dataset.

Note: Some assumptions can be checked before building the model, while others will be assessed after the model has been built.

In [35]:
```
# Create plot of pairwise relationships.
sns.pairplot(data)
```

Out[35]:
```
<seaborn.axisgrid.PairGrid at 0x20fe507e060>
```



Question: Is the assumption of linearity met?

Yes, the assumption of linearity is met, particularly for Google Ads Spend vs. Sales.

If you look at the scatter plot where Google_Ads_Spend intersects with Sales (in the bottom row, second column), the data points clearly form a tight, upward-sloping straight-line trend. This linear relationship confirms that as Google Ads spending increases, sales increase proportionally, validating the assumption required for linear regression.

Step 4: Model Building

Now that the relationship between the variables has been explored, we will build a simple linear regression model to examine the relationship between Google Ads expenditure and sales revenue.

The independent variable is `Google_Ads_Spend`, and the dependent variable is `Sales`.

In [37]:
```
# Build the simple linear regression model
ols_data = data[['Google_Ads_Spend', 'Sales']]
```

In [38]:
```
# Now we need to diplay the fisrt 10 rows
ols_data.head(10)
```

Out[38]:

	Google_Ads_Spend	Sales
0	38.34	179.15
1	57.36	242.27
2	55.24	205.24
3	59.49	194.11
4	43.37	162.59
5	54.27	229.71
6	105.13	358.29
7	77.90	235.32
8	17.39	122.94
9	75.44	283.13

In [39]:
```
# Write the linear regression formula.
# Save it in a variable.
ols_formula = "Sales ~ Google_Ads_Spend"
```

In [42]:
```
# Fit the model to the data.
# Save the fitted model in a variable.
model = OLS.fit()
```

In [41]:
```
# Implement OLS.
OLS = ols(formula = ols_formula, data = ols_data)
```

In [40]:
```
# Fit the regression model
model = ols('Sales ~ Google_Ads_Spend', data=ols_data).fit()
```

To build and run our regression model, we first define our formula in Cell [39] using ols_formula = "Sales ~ Google_Ads_Spend", which tells the model that we want to use Google Ad spend to predict sales. Then, in Cell [40], we pass this formula and our data into the Ordinary Least Squares function and call .fit() to train the model, allowing it to calculate the best-fit line and analyze the relationship between advertising spend and sales performance.

Step 4: Model Results

After building the simple linear regression model, we will examine the regression results to understand the relationship between Google Ads expenditure and sales revenue.

In [43]:
```
# Get summary of results.
model.summary()
```

Out[43]:

OLS Regression Results

Dep. Variable:	Sales	**R-squared:**	0.828
Model:	OLS	**Adj. R-squared:**	0.828
Method:	Least Squares	**F-statistic:**	2734.
Date:	Mon, 26 Oct 2026	**Prob (F-statistic):**	4.75e-219
Time:	23:59:14	**Log-Likelihood:**	-2755.8
No. Observations:	569	**AIC:**	5516.
Df Residuals:	567	**BIC:**	5524.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	81.7155	2.498	32.708	0.000	76.808	86.623
Google_Ads_Spend	2.0973	0.040	52.286	0.000	2.019	2.176

Omnibus:	0.225	**Durbin-Watson:**	1.852
Prob(Omnibus):	0.894	**Jarque-Bera (JB):**	0.224
Skew:	0.048	**Prob(JB):**	0.894
Kurtosis:	2.982	**Cond. No.**	121.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Project-Aligned Questions

1. **What is the y-intercept?**

Answer: The y-intercept is 81.7155.

Explanation: In the context of this data, it represents the baseline expected sales when Google Ads spend is zero.

2. **What is the slope?**

Answer: The slope coefficient for Google Ads Spend is 2.0973.

Explanation: This value tells us the rate of change in sales for every unit increase in advertising budget.

3. **What linear equation expresses the relationship between sales and Google Ads spend?**

Answer:Sales = 2.0973*(Google_Ads_Spend)+ 81.7155 Explanation: This formula allows us to predict overall sales revenue based on any given amount spent on Google Ads.

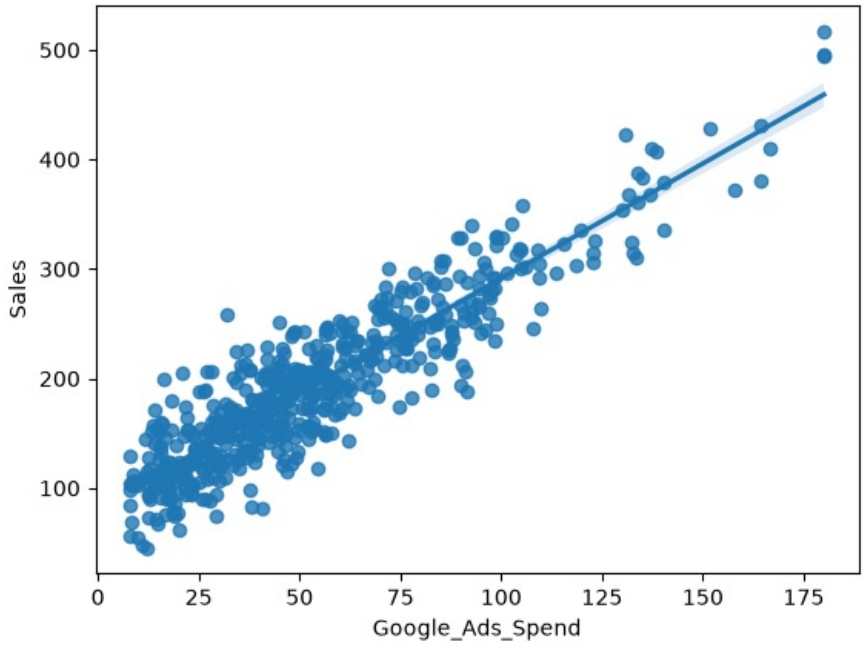
5. **What does the slope mean in this context?**

Answer: For every 1-unit increase in Google Ads spending, expected sales increase by 2.0973 units.

Explanation: This demonstrates a strong positive return on investment for money put into Google advertising.

In [44]:
```
# Plot the OLS data with the best fit regression line.
sns.regplot(x="Google_Ads_Spend", y="Sales", data=ols_data)
```

Out[44]:
```
<Axes: xlabel='Google_Ads_Spend', ylabel='Sales'>
```



Question: What do you observe from the preceding regression plot?**

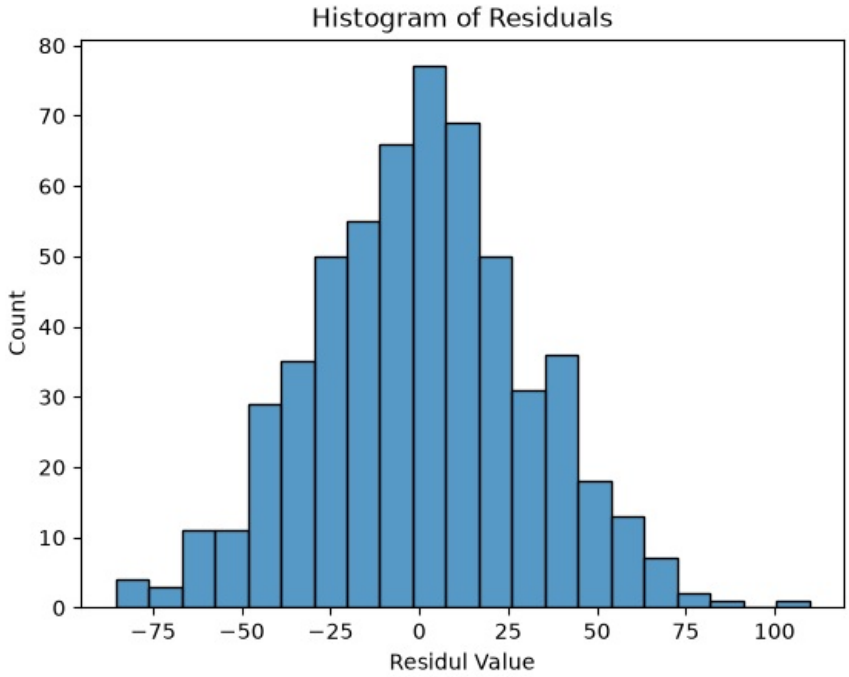
Strong Positive Linear Trend: The data points follow a clear, upward-sloping pattern along the blue regression line, confirming that higher Google Ads spend consistently corresponds to higher sales.

Tight Clustering with Slight Spread: The points are relatively close to the regression line for lower and middle budget ranges, though the spread widens slightly as spending and sales increase toward the higher end.

Positive Intercept & Slope in Action: You can visually see that even near zero ad spend, the baseline sales start around the y-intercept value (~81), and they scale up steadily as you move right along the Google_Ads_Spend axis.

In [45]:
```
residuals = model.resid
```

In [46]:
```
# Visualize the distribution of the residuals.
fig = sns.histplot(residuals)
fig.set_xlabel("Residul Value")
fig.set_title("Histogram of Residuals")
plt.show()
```

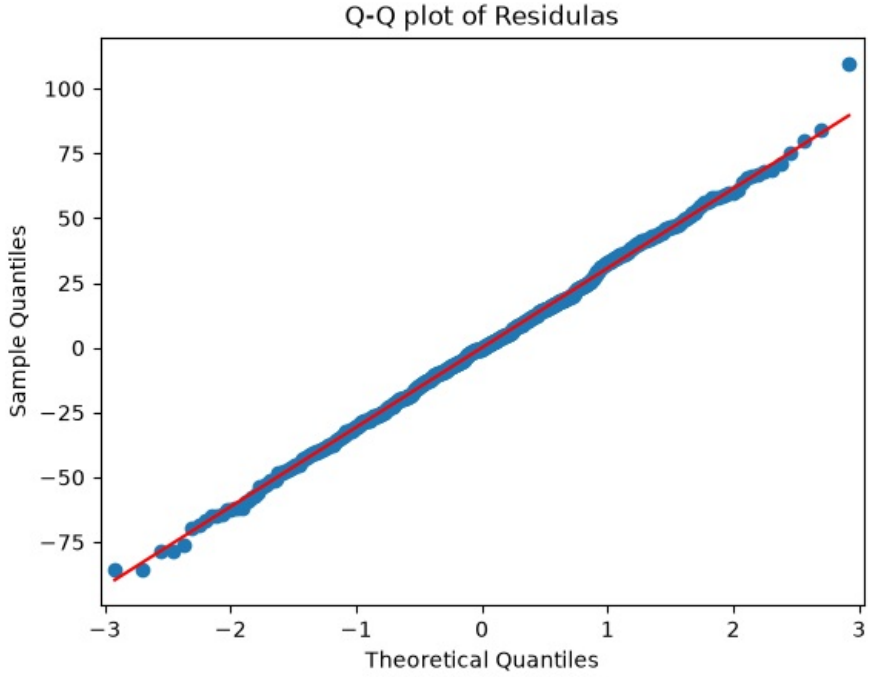


Question: Based on the visualization, what do you observe about the distribution of the residuals?** Normally Distributed / Bell-Shaped: The histogram shows a symmetric, bell-shaped distribution centered right around zero, which indicates that the residuals are normally distributed.

Centered at Zero: The highest frequency of residuals clusters tightly around the zero mark, meaning most of the model's predictions are very close to the actual values without systematic over- or under-prediction.

Low Spread and Few Outliers: The distribution tails off smoothly on both sides, confirming that extreme error values are rare. This satisfies the assumption of normality for linear regression residuals.

In [47]:
```
# Create a Q-Q plot.
sm.qqplot(residuals,line='s')
plt.title("Q-Q plot of Residulas")
plt.show()
```

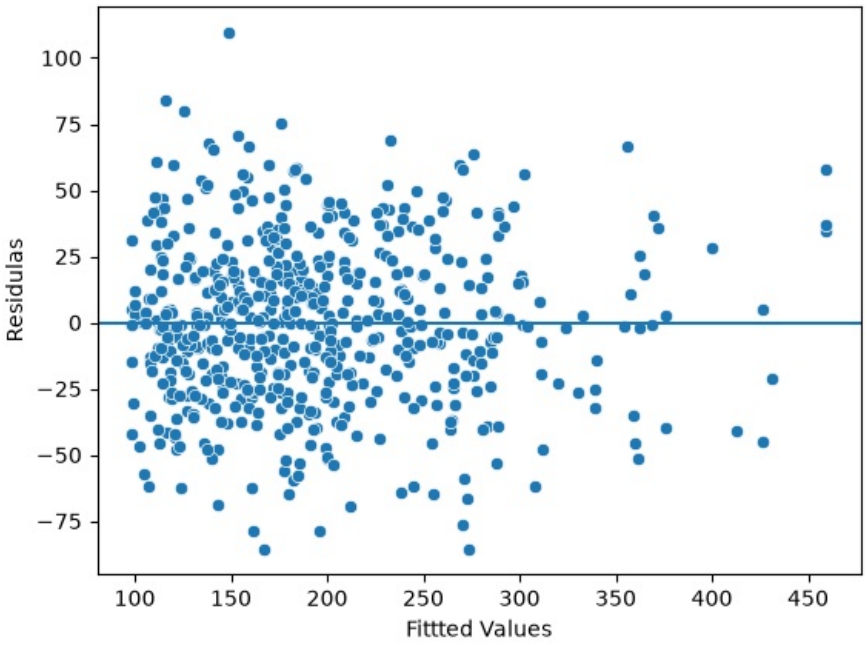


Question: Is the assumption of normality met?**

Yes, the assumption of normality is met for this linear regression model. As seen in the Q-Q plot above, the blue sample quantile points closely follow the red reference line, confirming that the residuals are normally distributed and satisfy this key regression assumption.

In [48]:
```
fitted_values = model.predict(ols_data["Google_Ads_Spend"])
```

In [49]:
```
# Create a scatterplot of residuals against fitted values.
fig = sns.scatterplot(x=fitted_values,y=residuals)
fig.axhline(0)
fig.set_xlabel("Fittted Values")
fig.set_ylabel("Residulas")
plt.show()
```



Question: Are the assumptions of independent observation and homoscedasticity met?

Yes, the assumptions of independent observations and homoscedasticity are met. As shown in the scatterplot above, the residuals are evenly and randomly distributed around the horizontal zero line across all fitted values without forming any funnel or fan shapes, confirming that the variance is constant and the observations are independent

Considerations

1. What are some key takeaways that you learned during this lab?

Strong Predictive Power: Google Ads spend has a strong, statistically significant positive relationship with sales, accounting for roughly 82.8% of the variance in sales (R^2 = 0.828).

Positive Return: Every single unit increase in Google Ads expenditure yields an estimated return of 2.09 units in sales.

Model Validation: Checking the residual plots, Q-Q plots, and scatterplots confirmed that the linear regression assumptions—including normality, homoscedasticity, and independence—were successfully met, validating the reliability of the model.

2. How would you present your findings from this lab to others?

Visualize the Trend: Show the scatterplot with the regression line to instantly demonstrate how sales scale up alongside advertising budgets.

Highlight the Equation: Share the linear model (Sales = 2.0973 * Google_Ads_Spend + 81.7155) to show how baseline sales and ad-driven growth are mathematically calculated.

Focus on Impact: Emphasize the p-value (0.000) to prove that the results reflect a genuine business trend rather than random statistical noise.

3. What summary would you provide to stakeholders?

Executive Summary: Google Ads is a highly effective, statistically proven driver of revenue growth, directly explaining the vast majority of our sales fluctuations.

Actionable ROI: Because the model shows a consistent positive return (~2.09 increase in sales per dollar/unit spent), continued investment in this channel is strongly justified.

Next Steps: Recommend exploring multiple linear regression by incorporating other marketing channels (like social media or video ads) to optimize overall budget allocation and check for potential channel saturation points.

In [ ]: